TEMCO LEARN

Psychrometric Tool

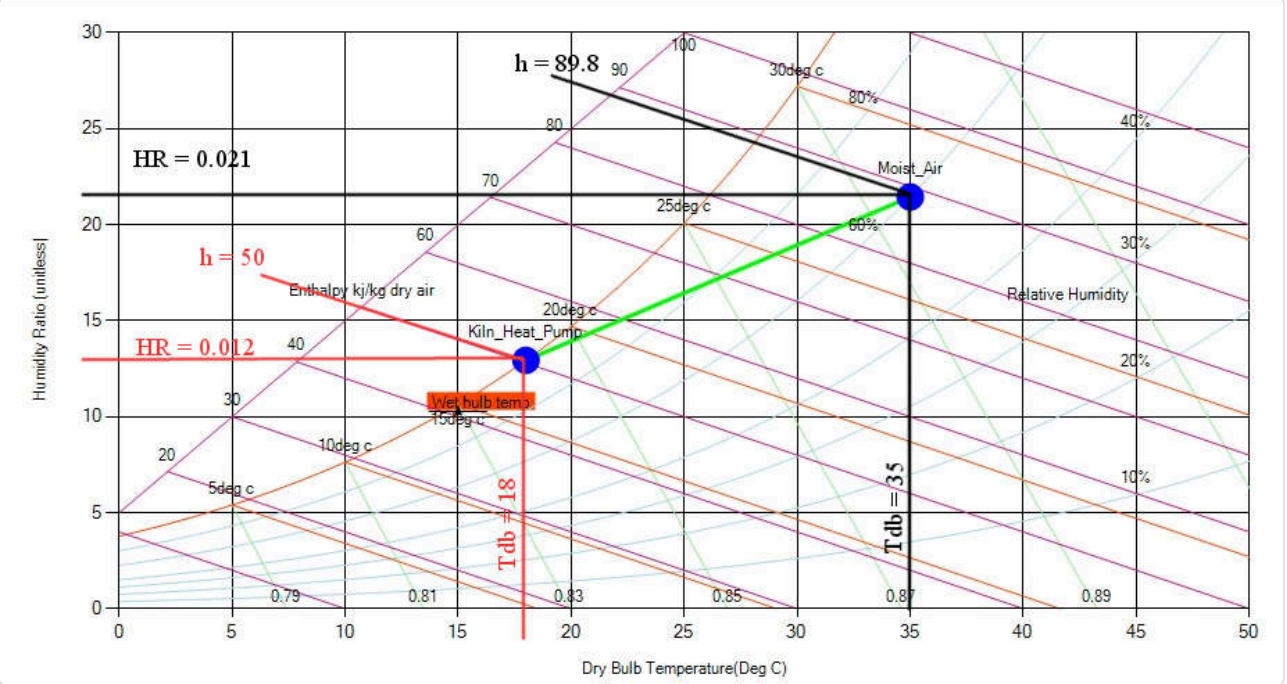
Numerical Example : Cooling & Dehumidification · built Nov 12, 2026

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Moist air at 35°C dry bulb temperature and 60% relative humidity enters the cooling coil of a dehumidification kiln heat pump system and is cooled to a temperature of 18°C. If the drying rate of 6 m3 of red oak lumber is 4 kg/hour, determine the W of refrigeration required.

Solution:

Following the steps used in the previous example of sensible heating we can get the general parameters.



Now:

ΔHR = 0.021-0.012

= 0.009 kg water / kg dry air

Wa = drying rate / ΔHR

= (4 kg water/h) /0.009

= 444.44 kg dry air / h

$\Delta h = (89.8 - 50)$ kJ/kg dry air

$= 39.8$ kJ/kg dry air

$q = (\Delta h)(Wa)$

$= [39.8 \times (kJ/\ kg\ dry\ air)]\ [444.44 \times (kg\ dry\ air/\ h)]$

$=17688.712$ (kJ/h)

$= 4913.531111W$

$= 4.913KW$